

EQUIPMENT SUBMITTAL

CRAH Units

Project: Meridian Data Centre Campus - Phase 1
Reference Section: 23 81 23
Package ID: SUB-238123-01-R0
Revision: R0
Reviewed Spec Revision: Rev 2024
Date: 01-01-2025

ACTIVE

Rev	Date	Description
R0	01-01-2025	Initial / Updated Submittal

Technical Parameters

Parameter	Claimed Value
Airflow	62390 m3/h
Overall Cabinet Width	2450 mm
Chilled Water Modulating Valve Type	2-way modulating control valve
Coefficient of Performance	5.3
EC Fan Power	5.6 kW
Entering Water Temperature	10 C
Fan Motor Power	7.5 HP
Fan Power Density	0.09 W/(m3/h)
Humidifier System Type	Infrared
Leaving Water Temperature	18 C
Operating Humidity Range	20% to 80% non-condensing
Refrigerant Type	R-407C
Sensible Cooling Capacity	250 kW
Water Temperature Delta-T	8 K

Clause ID	Requirement Summary	Response
23 81 23 Part 2.2.2.A	Minimum unit airflow rate shall be 64,320 m³/h at the design external static pressure. Airflow shall...	Comply
23 81 23 Part 2.2.4.C	Unit coefficient of performance (COP) shall be a minimum of 5.2. The vendor shall submit the COP val...	Comply
23 81 23 Part 2.3.2.B	Fan motor power shall be specified in electrical kilowatts (kW). Specifications of motor power in al...	Comply
23 81 23 Part 2.2.8	Where the CRAH unit incorporates a direct expansion (DX) trim cooling circuit, the refrigerant type ...	Comply
23 81 23 Part 2.3.5.A	CRAH unit shall be equipped with a 2-way modulating control valve for chilled water flow regulation....	Comply
23 81 23 Part 2.3.6.B	Humidity control shall be provided by an electrode steam generating humidifier integrated within the...	Comply
23 81 23 Part 2.3.5.C	Chilled water flow regulation shall utilise pressure independent control valves (PICV) to ensure con...	Comply
23 81 23 Part 2.4.1.A	CRAH unit overall cabinet width shall not exceed 2400 mm to ensure transport clearance through stand...	Comply
23 81 23 Part 2.2.1.A	Each CRAH unit shall provide a minimum sensible cooling capacity of 250 kW at the design conditions ...	Comply
23 81 23 Part 2.2.1.B	Design entering chilled water temperature shall be 10°C. The CRAH cooling coil shall be selected and...	Comply

Clause ID	Requirement Summary	Response
23 81 23 Part 2.2.1.C	Design leaving chilled water temperature shall be 18°C, resulting in a chilled water temperature dif...	Comply
23 81 23 Part 2.3.2.A	CRAH unit fans shall be electronically commutated (EC) type with variable speed drive. Fan motor pow...	Comply
23 81 23 Part 2.2.5	CRAH unit shall maintain operation across a relative humidity range of 20% to 80% non-condensing wit...	Comply

CERTIFICATE OF COMPLIANCE

This certifies that the equipment complies with standard testing requirements.

Issued by: Third Party Testing Laboratory